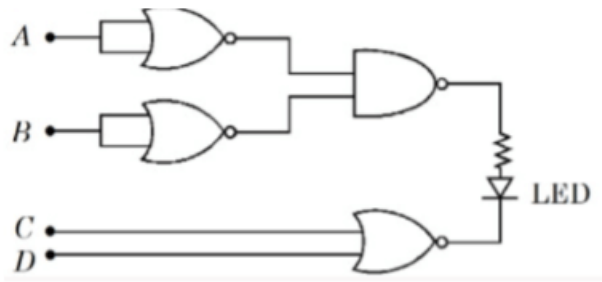


Question 1:

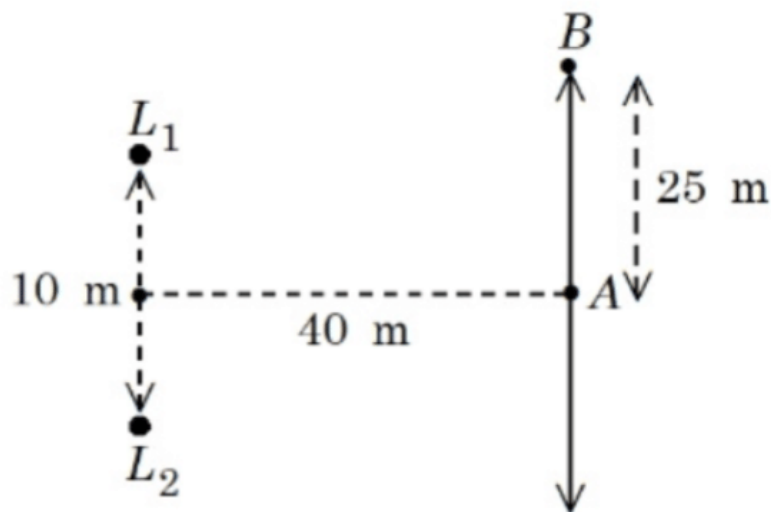
Find the correct combination of A, B, C and D inputs which can cause the LED to glow.



- (A) 0100
- (B) 1000
- (C) 0011
- (D) 1101
-

Question 2:

Two loudspeakers (L_1 and L_2) are placed with a separation of 10m, as shown in the figure. Both speakers are fed with an audio input signal of the same frequency with constant volume. A voice recorder, initially at point A, at equidistance to both loudspeakers, is moved by 25m along the line AB while monitoring the audio signal. The measured signal was found to undergo 10 cycles of minima and maxima during the movement. The frequency of the input signal is _____ Hz.



(Speed of sound in air is 324m/s and $\sqrt{5}=2.23$)

